Using Graphkit-Genco Grid Foundation Model to Recover the Aged Admittance Matrix

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Abstract

The grid admittance matrix is a critical, confidential input used in almost all grid calculations, including power flow, optimal power flow, and state estimation. However, this matrix changes over time as the grid ages. However, accurately tracking changes in line admittances can be challenging, particularly when direct measurements are unavailable or measurement coverage is limited. We propose using high-fidelity grid foundation models to estimate the aged admittance matrix. We use the GENCO model and assume that the grid operator has access to its nominal admittance matrix. We first show numerically that, using the nominal admittance matrix, the GENCO-estimated voltage magnitude and phase at buses are not biased and have symmetric error distributions around zero. However, using a fine-tuned GENCO model with the nominal admittance matrix to predict masked voltage amplitudes and phases from data samples generated with an aged admittance matrix yields skewed error probability density functions. We then propose a gradient-based procedure that estimates the aged admittance matrix by optimizing line impedances through a frozen GENCO model. We evaluate the accuracy of the estimated admittance matrix and demonstrate that it reduces the dispersion and systematic shift of voltage magnitude and phase-angle prediction errors.

1 Introduction

Grid Foundation Models (Grid-FM) offer a promising, transformative approach to addressing bottlenecks in the power industry's adoption of Artificial Intelligence. Grid-FMs use a generic representation of power-network operational data and topology, along with a physics-informed model architecture to overcome the limited transferability of developed models to adjacent applications and system configurations. They can be used across a broad range of applications and address limited data at each utility by enabling training of high-fidelity models [3].

This paper uses the Genco (Geometric Neural Corrective Optimization) foundation model, which provides a unified framework for optimal power flow (OPF), power flow (PF), and state estimation (SE) [5]. Genco is trained on diverse synthetic data generated by gridfm-datakit [6]. It uses a designed shared grid representation that could be used by three applications. Genco uses large-scale datasets with millions of PF and OPF scenarios and was evaluated on benchmark datasets.

GENCO deploys a physics-informed neural solver developed to accelerate steady-state transmission network analysis. OPF, PF, and SE are nonlinear and nonconvex and were traditionally solved with iterative numerical methods, each with its own solution approach. However, GENCO addresses these well-known power network problems simultaneously in a single shared model representation, using a common architecture that combines graph transformers, physics-constraint decoding, and iterative corrections guided by power balance residuals across heterogeneous graph neural network (GNN) layers. This design aims to improve computational efficiency while maintaining physical consistency across different network configurations and operating conditions, moving toward grid foundation models.

Training the GENCO model in a network-architecture-agnostic way and under different perturbations makes it robust to unseen networks and conditions. GENCO has strong potential for future use in power network studies. For example, the GENCO model can eliminate the need to install expensive measurement units by providing a relatively accurate estimate of the network state or, as we discuss in this paper, evaluating the consistency between grid operator data and measurements.

Aging in power networks refers to the progressive deterioration of equipment from thermal, electrical, mechanical, and environmental stresses over time, which can reduce reliability and change equipment characteristics. For example, in generators, mechanical and thermal stresses degrade winding insulation, whereas in circuit breakers, repeated operation erodes contact points and reduces switching reliability. Transmission lines also age from exposure to extreme weather and thermal stress, causing conductor corrosion, material deterioration, and drift in electrical impedance. In an aged transmission line, impedance characteristics increase from their nominal values; both resistance and reactance are higher than their nominal values. As a consequence of line aging, losses increase, so some energy is dissipated across the line during transmission.

The aging effects of transmission lines extend beyond power loss; they also create challenges for downstream power system analyses. Tasks such as SE and stability assessment rely on accurate transmission line impedance values. Uncertainty in line impedance can compromise assessments of operating-point feasibility and lead to unreliable evaluations of risks associated with proposed power-dispatch solutions. Continuous measurements and monitoring are also expensive. Therefore, it is paramount to find a cheap, viable method to estimate line impedance drift due to aging.

This study proposes using high-fidelity grid foundation models to first identify potential deviations from the nominal admittance matrix due to aging and subsequently estimate the aged admittance matrix. Specifically, we develop a GENCO-based algorithm to estimate aging-related changes in transmission line impedances. We investigate whether GENCO's learned representations can support accurate recovery of line parameters by iteratively optimizing impedance multipliers while keeping the model parameters fixed. The proposed approach reuses the existing forward model without training a dedicated inverse power-flow model or introducing an auxiliary parameter-estimation decoder.

The remainder of this paper is organized as follows. section 2 formulates the line parameter estimation problem, introduces the line aging model, and investigates GENCO's behavior under mismatched network conditions. section 3 presents the proposed impedance estimation algorithm, while section 4 evaluates its performance in recovering aged line parameters. Finally, section 5 concludes the paper.

2 Problem statement and solution approach

Previous studies have investigated conductor aging and its effects on electrical properties and reliability [4, 7]. Although maintaining accurate network parameters is important, updated information about transmission line conditions is not always available to network modeling and development teams. Detailed asset information may be restricted due to confidentiality or organizational data-sharing policies. Furthermore, direct inspection and offline measurement can be expensive, time-consuming, and may require temporary service interruptions. Although online parameter estimation techniques exist, they may depend on specialized measurement infrastructure and sufficient measurement coverage. As a result, network models may continue to rely on outdated nominal impedance values.

Transmission line parameter estimation using SCADA and PMU data has been addressed in previous studies [1]. However, we investigate an alternative approach using grid foundation models; here, the GENCO model. Instead of retraining the model, we propose a gradient-based algorithm that adjusts the nominal line impedances to better represent the aged network conditions.

Specifically, given the network topology, nominal line impedances, and available operational observations, we estimate updated resistance and reactance values by minimizing the discrepancy between observed PF quantities and those predicted by GENCO. The pretrained model parameters remain fixed, while line impedances are iteratively refined through backpropagation. The proposed approach aims to reduce reliance on direct impedance measurements and improve the accuracy of existing network models without requiring physical intervention, provided sufficient operational information is available.

A central challenge is ensuring that the estimated impedances approach their actual aged values rather than merely minimizing PF prediction error, since different impedance configurations can produce similar

network responses. To mitigate this issue, we conduct experiments across multiple aging scenarios generated using different random seeds to identify an appropriate early stopping criterion.

2.1 Line aging model

To investigate the effects of aging on GENCO's predictions, we generated an aged IEEE-118 network from its nominal configuration. Weibull distributions have been employed in previous studies to model conductor aging and failure probabilities [7, 2]. Following this approach, we assigned synthetic ages to transmission lines using a right-truncated Weibull distribution with shape parameter $k = 2.5$, scale parameter $\lambda = 40$ years, and maximum age $A_{\max} = 60$ years. The probability density function is given by:

$$f_A(a) = \frac{k\, a^{k-1} e^{-(a/\lambda)^k}}{\lambda^k \left(1 - e^{-(A_{\max}/\lambda)^k}\right)}, \quad 0 \le a \le A_{\max}. \tag{1}$$

where a denotes the transmission line age. The resulting age distribution is illustrated in Figure 1.

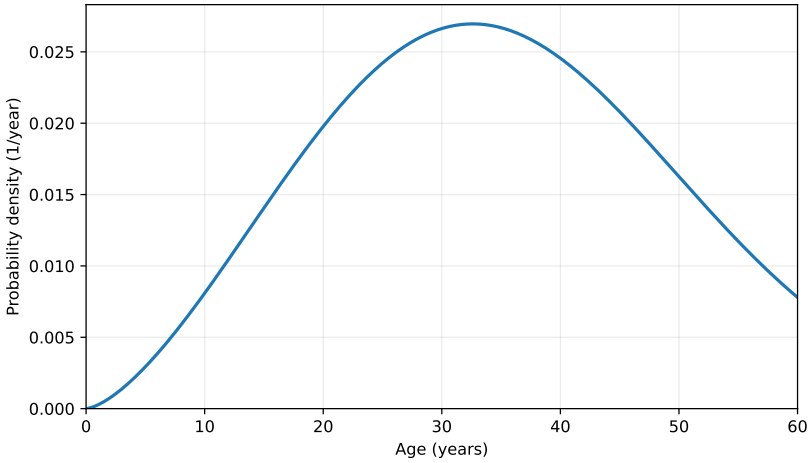


Figure 1: Right-truncated Weibull distribution used to model transmission line aging.

The assigned ages were mapped to impedance increases ranging from 0% to 50% using:

$$D_i = 0.5 \left(\frac{A_i}{60}\right)^2, \tag{2}$$

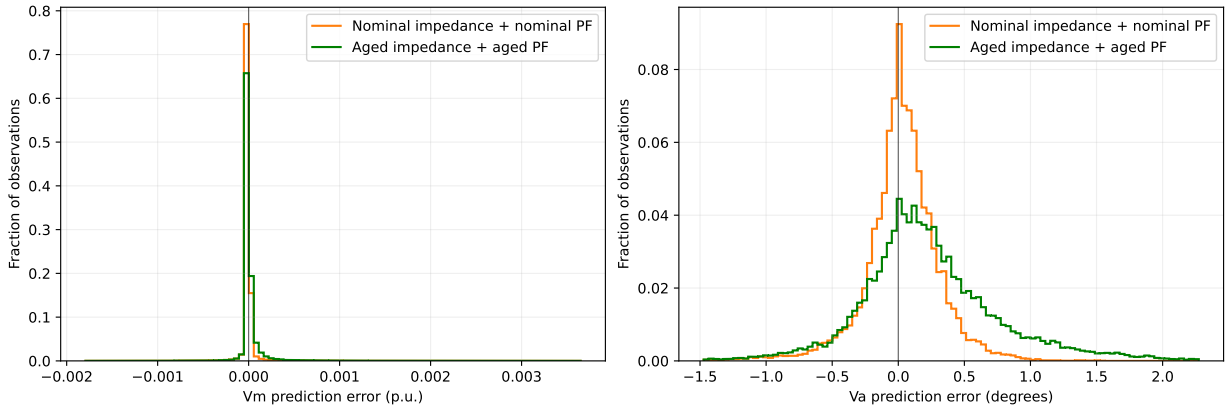
where A_i represents the age of line i. We multiplied resistance and reactance independently by the factor $(1 + D_i)$. These parameters represent a synthetic aging scenario rather than an empirically calibrated degradation model.

2.2 Solution approach using foundation models

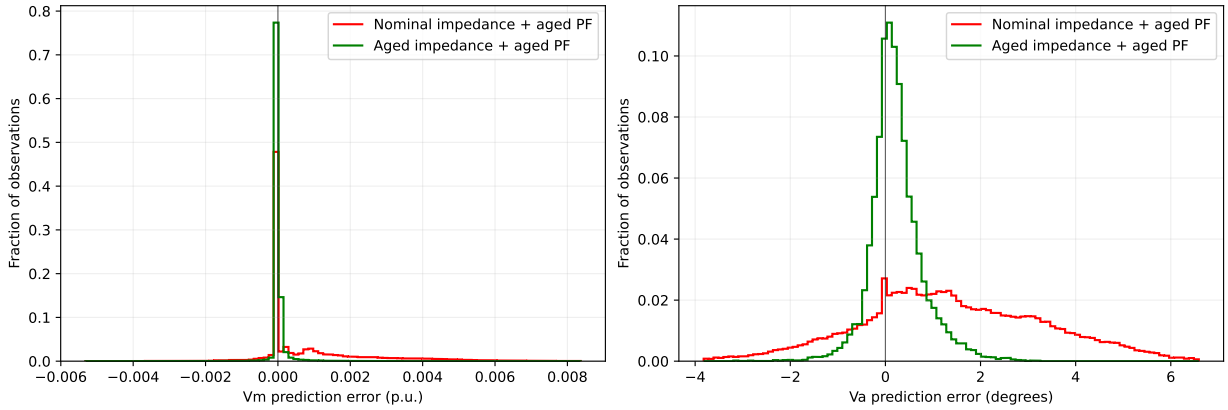
We rely on the observation that GENCO's prediction errors are approximately centered around zero when the supplied admittance matrix matches the network underlying the PF observations. We then investigate how line impedance mismatches due to aging affect GENCO's performance when predicting network states from SCADA-like PF observations generated using the aged admittance matrix.

Case 1: Matched admittance and SCADA-like PF observations

Under matched conditions, aging increases GENCO's prediction errors, particularly for voltage phase angles (V_a), whose error distribution becomes wider. However, the signed prediction errors remain approximately centered around zero, with no pronounced systematic shift, as shown in Figure 2a. Across seeds 0–4, the mean signed V_a error ranged from 0.0240° to 0.0438° under nominal matched conditions and from 0.0237° to 0.2336° under aged matched conditions. These results suggest that GENCO's prediction errors remain



(a) Matched network conditions: nominal and aged impedances.



(b) Matched and mismatched impedance conditions using aged PF observations.

Figure 2: GENCO prediction-error distributions for the IEEE-118 network under (a) aged and (b) nominal impedance conditions.

Table 1: Impedance estimation errors for IEEE-118 across seeds 0–4. Initial and final values correspond to epochs 0 and 250, respectively. RMSE values are in p.u.

	Resistance RMSE			Reactance RMSE		
Seed	Nominal	Estimated	Reduction (%)	Nominal	Estimated	Reduction (%)
0	0.007851	0.005306	32.42	0.030203	0.018199	39.75
1	0.007062	0.005127	27.41	0.025700	0.015182	40.93
2	0.008474	0.005388	36.42	0.031549	0.018433	41.57
3	0.008010	0.005433	32.17	0.029272	0.017257	41.04
4	0.007895	0.005576	29.38	0.030815	0.017111	44.47

approximately centered around zero when the supplied admittance matrix matches the network underlying the PF observations.

Case 2: Mismatched admittance and SCADA-like PF observations

In the second experiment, we investigate GENCO's response to mismatched network parameters. Specifically, we use the nominal admittance matrix to predict network states from PF observations generated using the aged admittance matrix. As shown in Figure 2b, this mismatch produces a pronounced positive shift in V_a prediction errors. Voltage magnitude (V_m) errors also increase, although the systematic shift is more evident in V_a. Across five aged networks generated using different random seeds (0–4), the mean signed V_a prediction error ranged from 1.254° to 1.866°. Notably, even the smallest mean error under mismatched conditions exceeds the largest mean error observed under matched conditions by more than a factor of five.

These observations suggest that GENCO's prediction residuals can serve as indicators of line impedance mismatch. Specifically, a systematic shift in the mean prediction error relative to a matched-condition baseline may indicate that the nominal admittance matrix no longer accurately represents the aged network. This motivates using GENCO as a surrogate PF model to detect potential impedance drift by comparing its predictions with available operational observations.

3 Impedance Estimation Algorithm

To estimate aged line impedances, we propose a gradient-based optimization method that uses GENCO as a surrogate PF model. As illustrated in Figure 3, GENCO's parameters remain frozen throughout the process, while the line resistances and reactances are treated as learnable parameters.

Rather than directly updating the impedances, independent multipliers are introduced for each eligible transmission line (bus-to-bus lines):

$$\hat{R}_l = \alpha_l R_l^0, \qquad \hat{X}_l = \beta_l X_l^0, \quad \alpha_l, \beta_l \in [1, 1.5], \tag{3}$$

where R_l^0 and X_l^0 represent the nominal line parameters. The estimated impedances are converted to admittances and supplied to GENCO alongside the aged network's operating conditions.

The optimization minimizes the mean squared error between GENCO's predicted and actual bus voltage magnitudes and angles. Gradients are backpropagated through the frozen model to update only the impedance multipliers. We use Adam with a learning rate of 3×10^{-3}, accumulating gradients over the entire dataset before each update. The multipliers are subsequently projected onto their predefined bounds.

To select the optimization hyperparameters, five independent experiments were conducted using different impedance perturbations generated with random seeds 0–4. We used the convergence behavior of resistance and reactance estimation errors, shown in Figure 4, to determine an appropriate stopping criterion and prevent overfitting.

4 Results

The proposed method is evaluated in terms of impedance estimation performance and the resulting improvement in PF predictions.

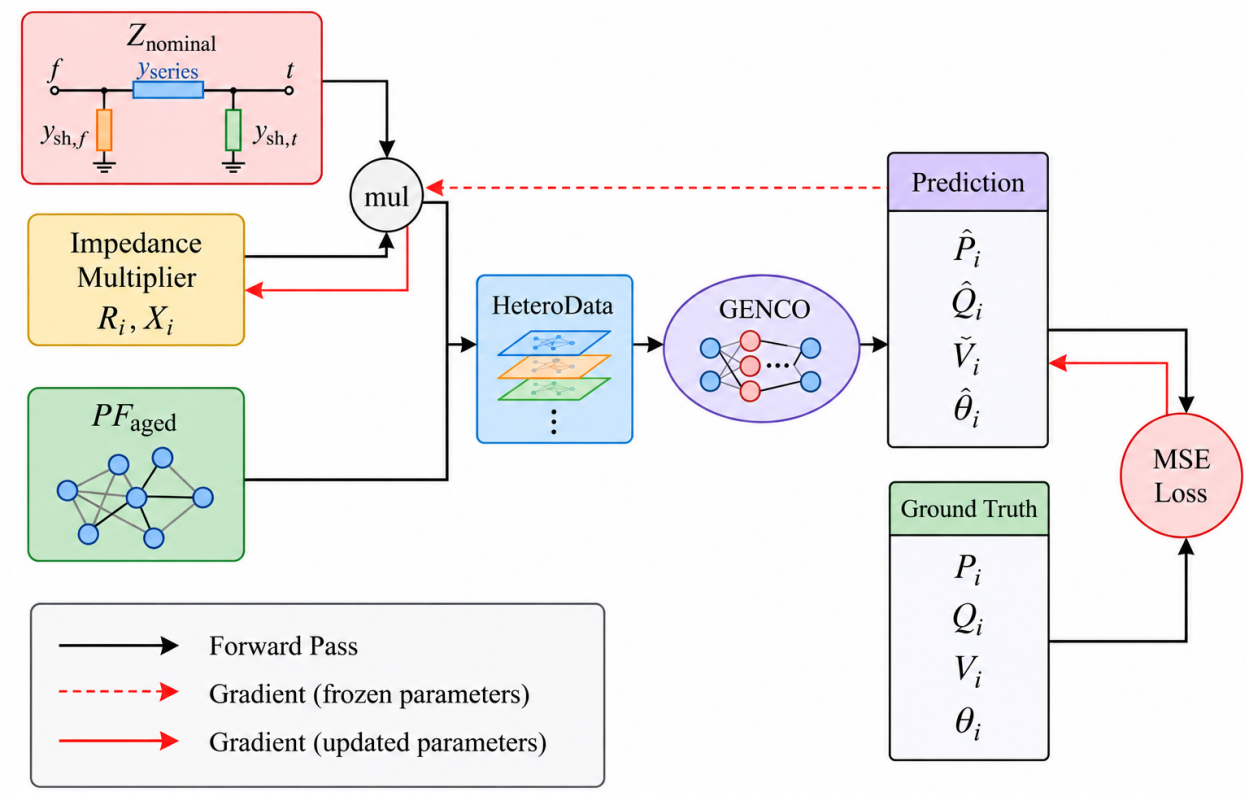
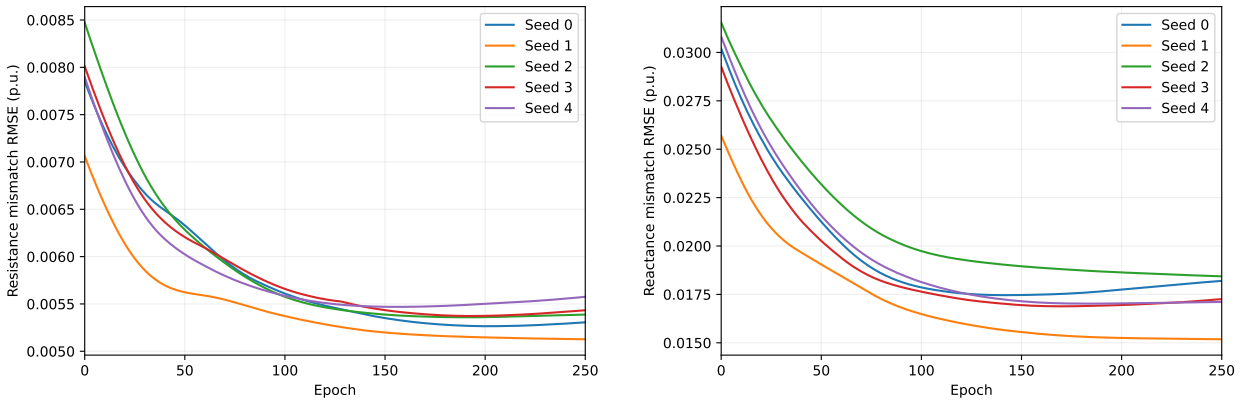


Figure 3: Proposed GENCO-based impedance estimation framework. The model parameters remain frozen while gradients are propagated to the learnable impedance multipliers.



(a) Resistance mismatch RMSE

(b) Reactance mismatch RMSE

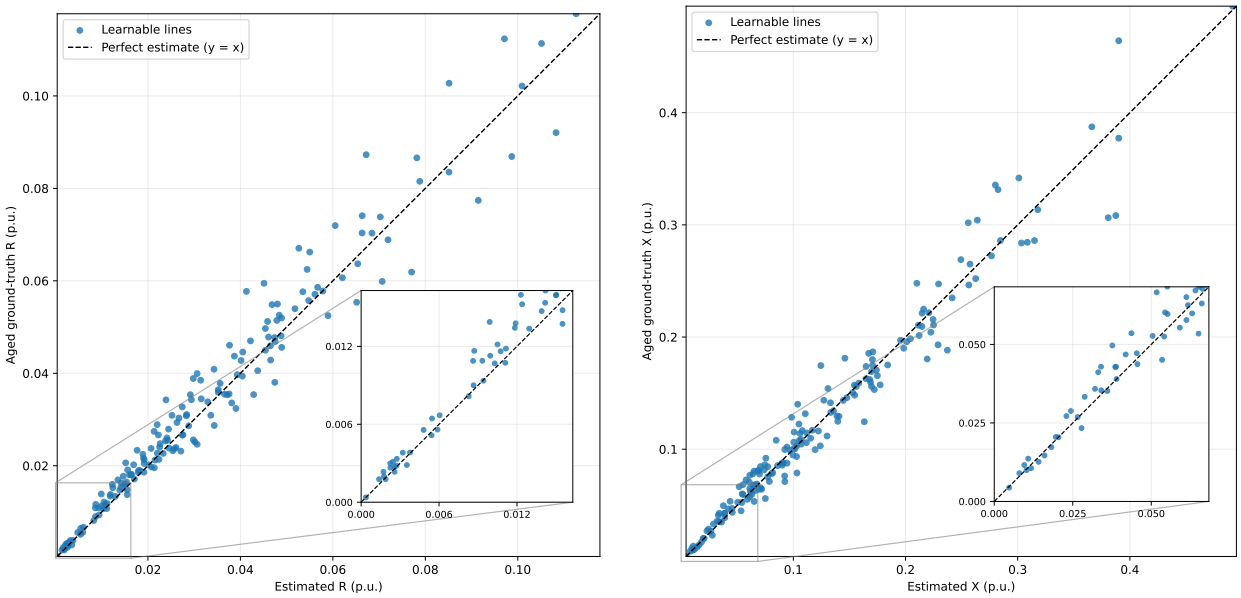
Figure 4: Evolution of impedance estimation errors during optimization across five independent experiments (seeds 0–4).

First, the optimization dynamics presented in Figure 4 demonstrate an overall reduction in impedance mismatch. For seed 0, the resistance RMSE decreases from 0.007851 to 0.005306 p.u., corresponding to a 32.42% reduction. Similarly, the reactance RMSE decreases from 0.030203 to 0.018199 p.u., representing a 39.75% improvement. Table 1 summarizes the nominal and estimated RMSE values across all five seeds. Resistance RMSE reductions range from 27.41% to 36.42%, while reactance RMSE reductions range from 39.75% to 44.47%.

Figure 5 compares the estimated resistance and reactance values against their aged counterparts for the experiment with seed 0. Most points are concentrated near the ideal $y = x$ line, indicating that the estimated parameters generally approximate the aged impedances. Nevertheless, some deviations remain, particularly for lines with larger impedance values.

Finally, Figure 6 compares GENCO's PF prediction errors under three conditions: (i) using the aged admittance matrix, (ii) using the nominal admittance matrix, and (iii) using the estimated admittance matrix. In all cases, the PF reference values correspond to the aged network.

Using the nominal admittance matrix introduces a noticeable shift and dispersion in the prediction errors, particularly for V_a. After impedance estimation, the error distributions become substantially narrower and are again concentrated around zero. Although the predictions do not fully match those obtained using the aged impedances, the results demonstrate that the proposed method can substantially mitigate the effects of unknown impedance drift.



(a) Resistance estimation

(b) Reactance estimation

Figure 5: Scatter plots comparing estimated and aged line impedances for the IEEE 118-bus network (seed 4).

5 Conclusion

We investigate using the grid foundation model GENCO to estimate unknown transmission line impedances affected by aging. Despite the simplicity of the proposed gradient-based approach, we showed that a pre-trained forward PF model can address the impedance drift problem of the admittance matrix. The estimated impedances agreed meaningfully with the aged line parameters and substantially improved PF predictions under impedance mismatch.

More importantly, these findings suggest that GENCO's capabilities extend beyond those of a conventional PF regressor. The informative gradients with respect to line impedances indicate that the model has

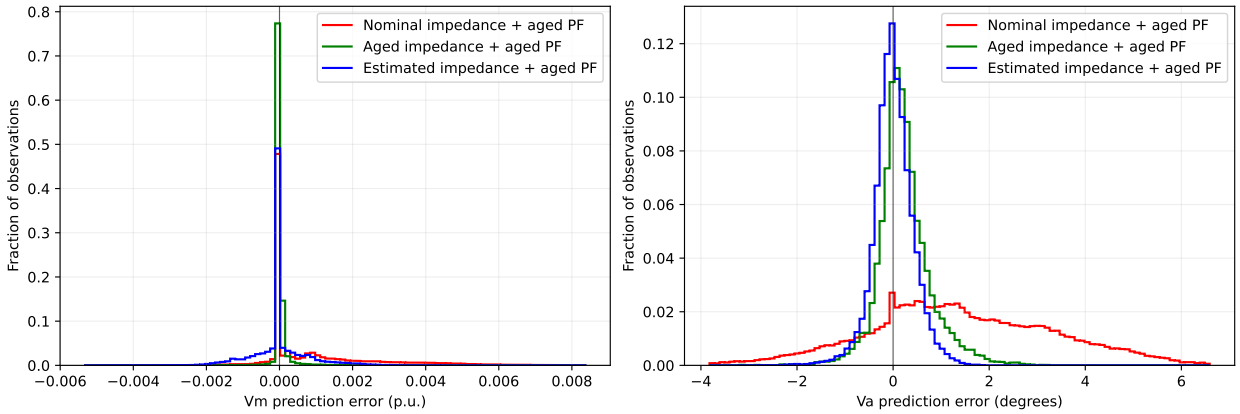


Figure 6: Distribution of power-flow prediction errors for voltage magnitude (left) and voltage angle (right). Three admittance configurations are compared: actual aged (orange), original (red), and estimated (green).

learned useful relationships between network parameters and operating conditions. Although this does not establish full model interpretability, it provides a promising direction for investigating the physical significance of its learned representations. Future research should further explore GENCO's potential for aged PF problems, parameter identification, and gradient-based model interpretability.

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